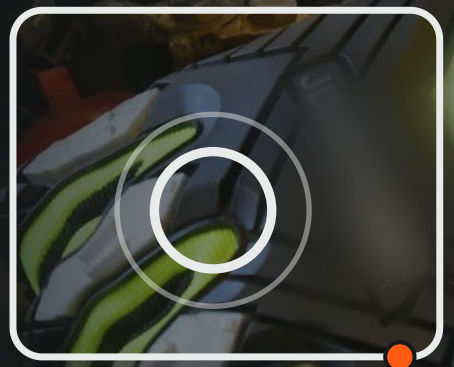
praxis

STEP 4/9

Instructions that can finally see how you're doing

Software for smart glasses that coaches new maintenance technicians, step by step, from the manuals a company already has.



Seat the clamp past the hose bead

STARTUPS @ HARVARD · 20 OCT 2026 · MONISH SARAVANA

CONCEPT MOCK · STOCK PHOTO

THE PROBLEM

Every new maintenance hire consumes scarce senior-technician time

WHAT ONE NEW TECH COSTS TO RAMP UP

$10–15k

to hire and ramp up one maintenance technician, counting onboarding time and lost productivity

HOW OFTEN

~255,000

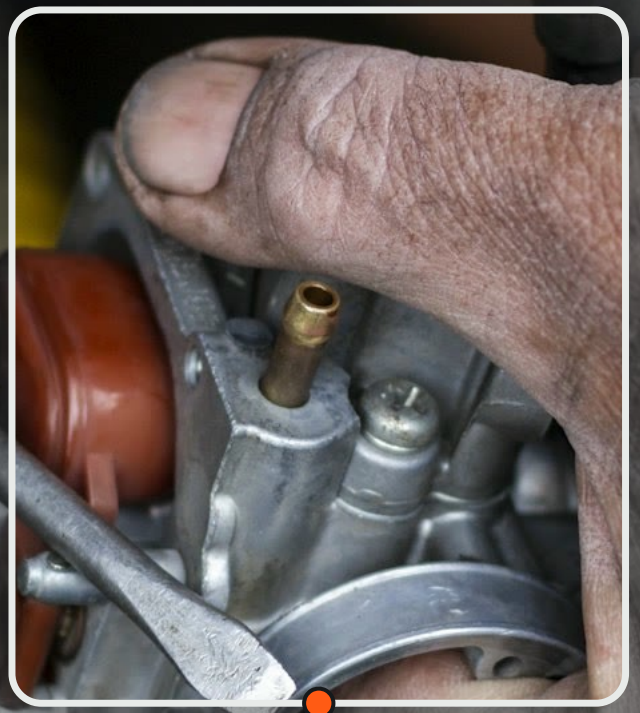
US maintenance and repair openings a year. Every new hire pulls a senior tech off the floor to teach them.

Why it's senior time: manuals and videos can't see what a new tech is doing, so someone experienced has to stand next to them. In a survey of 35 repair stations, new hires took 9–14 months to pay for themselves.

THE PRODUCT

Upload the manual. Praxis coaches the job through smart glasses

LOAD	the manuals a company already has. Each procedure becomes a task model
COACH	through smart glasses: it knows the step you're on and says the next one
VERIFY	important steps, catch mistakes, and ask for a better view instead of guessing



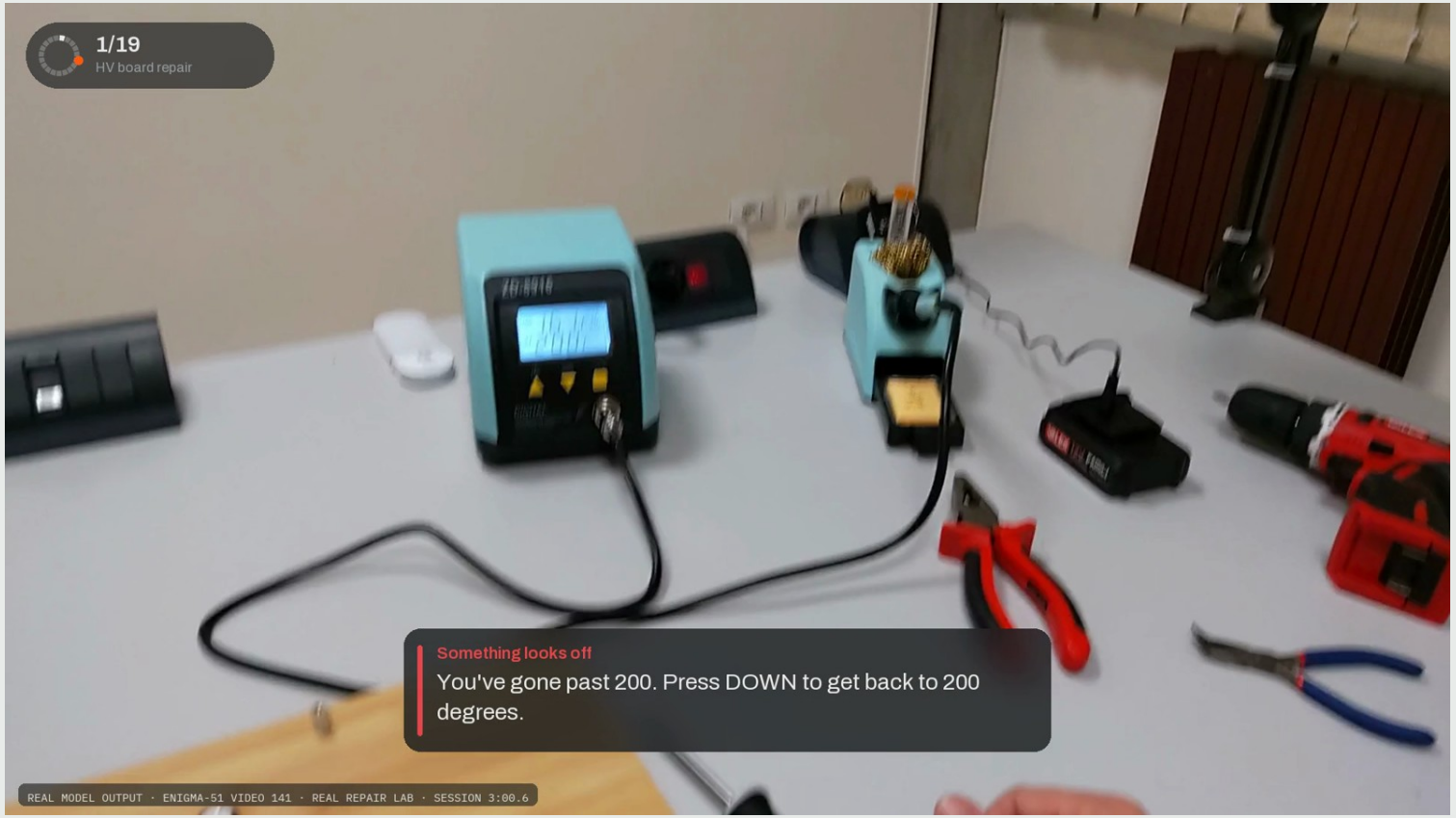
STEP 3/8

Loosen the four float-bowl screws a half turn each

CONCEPT MOCK · STOCK PHOTO

PROOF IT'S REAL

Real repair footage, run through the live engine. Every card on screen is the model's own output



SESSION

16:40

live run on first-person footage of a real high-voltage board repair

TRAINING ON THIS JOB

None

It started from the lab's written procedure

MODEL COST

$0.17

for the whole session

ENIGMA-51 PUBLIC DATASET · VIDEO 141 · INDUSTRIAL REPAIR LAB

HOW IT WORKS

The frontier model is one swappable part. The tracking engine around it is ours.

OURS

Manual → task graph

a company's own procedure → steps, timing, checks

OPEN MODEL

Camera + perception

first-person video turned into features by open models (SigLIP 2, V-JEPA 2)

OURS · TRAINED

Temporal state tracker

Bayesian semi-Markov model + learned event models: which step you're on, for how long, and what comes next

belief over the steps

OURS

Policy → coach

says the next step, verifies, flags a mistake, or asks for a better view

question

evidence

RENTED · SWAPPABLE

Verifier: a frontier model

asked only when something happens: ~1 in 7 frames

Ours · trained on labeled recordings

Ours

Open model

Rented · swappable in one line

FULL ARCHITECTURE · APPENDIX A4

MEASURED · 48 HELD-OUT FIRST-PERSON VIDEOS

Same frontier model. Better system.

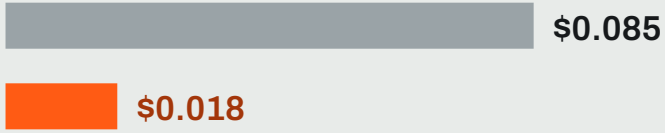
Baseline: the same frontier model (Claude Opus), asked about the video every 5 seconds, with no memory of the task.

CALLS TO THE FRONTIER MODEL

8× fewer

routine checks go to a fast, cheap model

MODEL COST PER MINUTE OF VIDEO



5× cheaper

about $1 per coached hour

STEP TRACKING ON VIDEOS IT NEVER SAW

Tracks the current step more accurately than the baseline (+13% relative), despite 8× fewer frontier calls.

Better

raw numbers: appendix A1

WHEN IT CAN'T SEE, IT SAYS SO

NEED A BETTER VIEW

Can't see the bearing cap.

Lean in and hold still for two seconds.

EXAMPLE CARD · CONCEPT

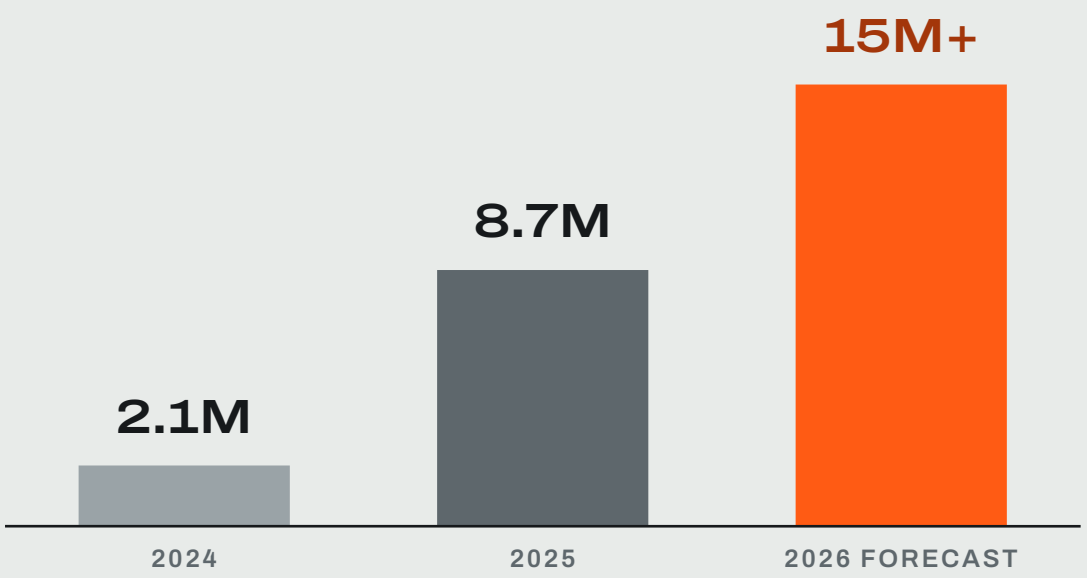
Praxis: never confidently wrong
on blurred or blocked views. It asked for a better view on 77–98% of them.

Baseline: confidently wrong on up to 73%
It has to answer, so it names a step anyway.

WHY NOW

The hardware layer just became a platform

AI GLASSES SHIPPED A YEAR, WORLDWIDE



AND THE PLATFORMS OPENED UP

2025	Meta opens a developer preview: outside apps can use its glasses' camera
SEPT 2026	Meta's toolkit reaches 1.0: camera, microphones and speakers for outside apps
FALL 2026	Google says its first Android XR glasses arrive: audio, with a camera
Q1 2027	Viture + NVIDIA ship Helix: safety glasses that coach workers on SOPs in real time

Three years ago we would have had to build the glasses.
Now we build the intelligence that runs on them.

COMPETITION

We don't need to win the glasses war

THEIR LAYER · HARDWARE, SENSORS, DISTRIBUTION, A GENERAL ASSISTANT

Meta
Ray-Ban, Oakley, Display · toolkit 1.0

ADAPTER

Google + Samsung
Android XR glasses

ADAPTER

Viture + NVIDIA
Helix glasses · XR AI

ADAPTER

Next devices
each one is one more adapter

ADAPTER

OUR LAYER · PRAXIS

Maintenance procedure intelligence

Manuals → task graphs · step-by-step state · verification · when to speak · records

EVERYONE ELSE

Procedure software
MaintainX, Tulip, Augmentir

Puts the steps on a screen with a checkbox. Can't see whether a step was done right.

Fixed cameras
Retrocausal, Invisible AI

Watch one station on a line. Can't follow a tech around a machine.

General assistants
Meta AI, Gemini Live

Answer about what's in view. No procedure, no step state, no records.

Viture + NVIDIA · closest
Helix, ships Q1 2027

First-person, real-time SOP coaching, sold with its own glasses. We'd rather run on Helix than fight it.

A $306M first wedge, at a price a maintenance manager can approve

THE FIRST WEDGE, PER YEAR

~255k new maintenance techs a year × $200 a month per seat × 6 months of guided onboarding ≈ **$306M** a year

WHY A MAINTENANCE MANAGER SAYS YES

$200/mo per new-tech seat ≈ **1 hr/week** of a senior tech's time (about $44 an hour, loaded)

Save a senior tech one hour a week and the seat pays for itself.

Later, on the same engine: every US maintenance tech (~$1.8B a year), then every worker who follows written procedures (~$9.2B). Math: appendix A9.

HOW WE CHARGE

Paid pilot first
with the success metric agreed before we start

$200 a month per active new-tech seat
billed annually, unlimited sessions

The seat moves to the next new hire
once a tech works alone, about six months in

Glasses are shared site equipment
use supported glasses you own, or buy a small pool. We don't resell hardware

Optional
procedure setup; later, skill records for the whole team

GO-TO-MARKET

We sell to one maintenance manager at a time, starting with a paid pilot

Email + call maintenance managers → Demo on their own manual → Paid pilot, success metric agreed up front → Annual deployment: a seat per new hire

WE ARE HERE

FIRST SIGNAL · OUTREACH STARTED OCT 17

70 emails to maintenance employers, training programs and associations, 68 delivered.
2 replies so far: early signal, not customers.

> **“The hardest AND most expensive part is training new techs …”**
> HOSPITAL STERILE PROCESSING ASSOCIATION

> **“We would be open to an experiment like this.”**
> FIBER-OPTIC TRAINING SCHOOL · ABOUT ONE LAB EXERCISE

SECOND LOOP · TRADE SCHOOLS

1. Students learn hands-on skills with Praxis
2. They bring it to their first employer
3. That employer buys seats for its next hires

Also how we'd collect consented training footage.

WHY ME

I've built real-time vision on messy real-world video before, and I build fast

PRAXIS, SO FAR

4 days

from first commit (Oct 17) to a tracking engine, a company backend, a live app and a 48-video evaluation. Solo, 130+ commits.

LIPTRAC	Real-time lip reading from ordinary video: 60,000+ frames, 12 testers, provisional patent filed. Temporal vision under changing light and camera angles.
ROBOTICS	7 years of competitive robotics. I learned how hard it is to mentor someone without seeing what they see.
AERIAL VISION	Autonomous path planning for oil-cleanup boats from aerial imagery.
NOW	Harvard College · solo founder of Praxis

THE LONG GAME

Coaching is the wedge. What compounds is maintenance data, models and skill records

NOW

Coach + verify new techs

Paid pilots with maintenance teams, running from each customer's own manuals.

NEXT

Maintenance-specific models

With customer permission, sessions can become labeled training data. Our models get cheaper to run and better at maintenance.

THEN

Verified skill records

Planned: certification partners run their hands-on checks on Praxis. Techs carry the record to their next shop.

LONG-TERM OPTION

Physical AI

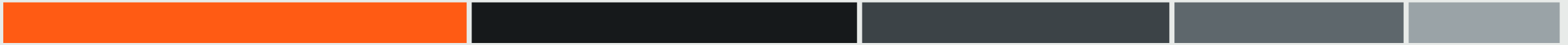
The same step-verified data, and a checker that knows whether a step succeeded, could help train and check robots.

WHY IT COMPOUNDS

More sessions → more consented, labeled data → better, cheaper models → better coaching → more sessions

THE ASK

$10k turns a working prototype into a first paid pilot on real glasses



$3,000

GPU + model training

Retrain the step tracker, event detector and local checker on real maintenance footage

$2,500

Devices + capture

Validate Praxis end to end on supported glasses at a real site

$2,000

Data

Pay techs and trade-school students to capture and label ~80 hours of real work, with consent

$1,500

Pilot inference

About 1,400 coached hours at today's measured cost

$1,000

Company + legal

Formation and the contracts a first paid pilot needs

TRUE BY APRIL

Models retrained on real maintenance footage

Praxis running end to end on glasses at a real site

A first paid pilot, with a success metric agreed up front

Q&A SLIDES

Appendix

A1 Benchmark: raw numbers + method

A2 Dataset, splits, denominators

A3 Learning curves

A4 Full architecture

A5 Live session internals + latency

A6 The backend a company uses

A7 What we train, what we rent

A8 Competitors in detail

A9 Market math + assumptions

A10 Unit economics + pricing risk

A11 Hardware, adapters, glasses + seats

A12 Data rights + privacy

A13 Safety model

A14 Risks + what we do

A15 Why not consumers first

A16 Founder details

A17 Long game: revenue, certs, robotics

APPENDIX · A1

Benchmark: the raw numbers and the method

SYSTEM (SAME 48 HELD-OUT CLIPS, TASK GIVEN)	STEP TRACKING	FRONTIER (OPUS) CALLS	COST / VIDEO-MIN
Baseline: Claude Opus every 5 s, no task state	48%	2,817	$0.085
Praxis: fast tier (Jev) + Opus escalation	54%	351 (+2,679 fast-tier)	$0.018
Praxis, fast tier only (no escalation)	53%	0	$0.001
Praxis + our local student on routine checks	52%	not broken out	$0.008
Our local layers only, no model calls	41% (49% retrained)	0	$0

Data. 48 held-out first-person clips (235 min) from CaptainCook4D, a public procedure dataset (cooking): 8 procedure types, 302 annotated steps, 170 annotated mistakes. Labels are the dataset's. Train and test never overlap.

Metric. Step tracking = share of frames where the tracked step matches the label. One run per system; no confidence intervals yet.

Mistakes (strict, judge-verified). Praxis alerts named a real annotated error 4 of 17 times; the baseline 2 of 3. Both catch a small share of the 170.

Degraded views (60 checks per condition). Praxis asked for a better view on 77–98% and was never confidently wrong; the baseline was wrong at ≥50% confidence on 12–73%.

Repair Q&A (ENIGMA-51). 11 of 15 technician questions answered correctly from the uploaded procedures, 2 partly, 2 wrong, blind-judged.

Training data, stated plainly

Train and test never overlap. Everything learned uses only the dataset's official train-split recordings

Small on purpose. A weekend-scale proof: the step-change detector trained on 24 recordings, the step models on 3 per procedure, the student on checks from 30

Tested on 48 held-out clips: 235 min, 302 annotated steps, 170 annotated mistakes, plus procedure types never seen in training

Design choices were made on train-split recordings only, never on the test clips

Labels are the dataset's, not ours. One run per system, no confidence intervals yet

LEARNING CURVES

Step models: 30.7% with 1 recording → 43.4% with 3 → 50.7% with 9 (the 3 procedures with 9)

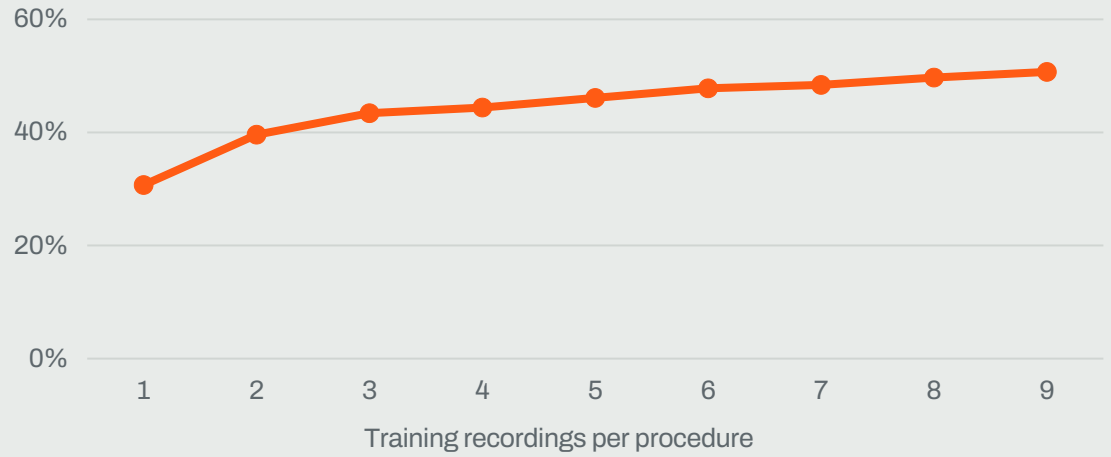
Student: 62.6% with 4 recordings → 72.5% with 30; the frontier teacher is 79.7%

Detector: noisy, no clear trend yet. Not claimed

Both learning curves: step models and the local student

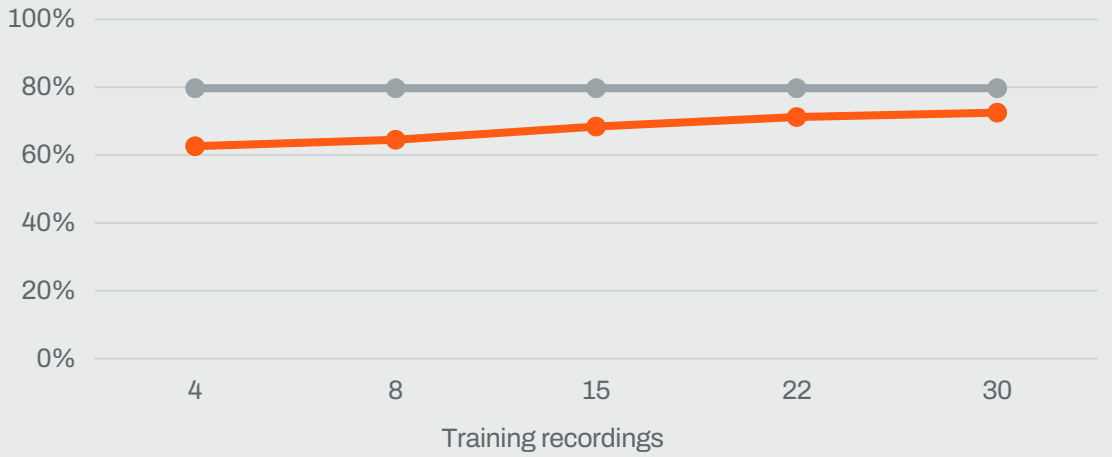
30.7% → 50.7%

recognizing the current step, on videos it never trained on



62.6% → 72.5%

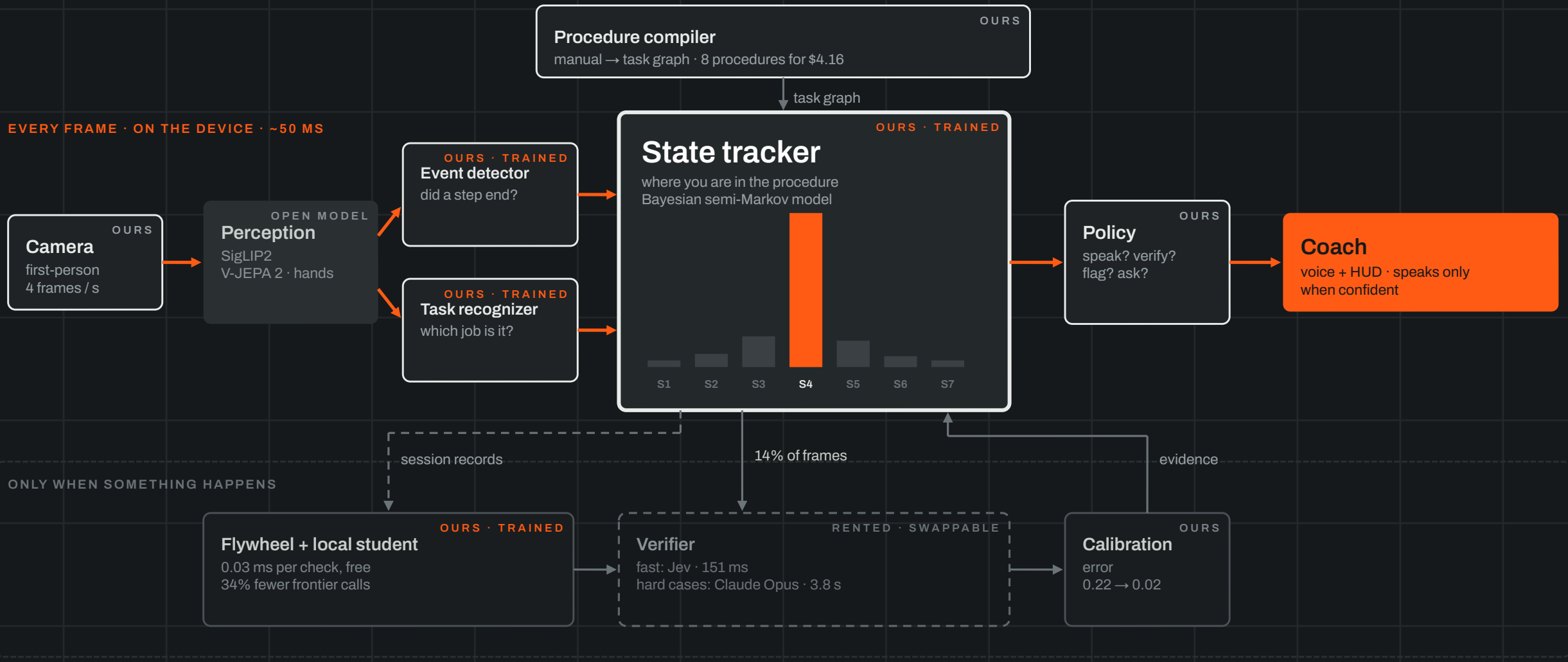
checking a step is done · grey: the frontier model, 79.7%



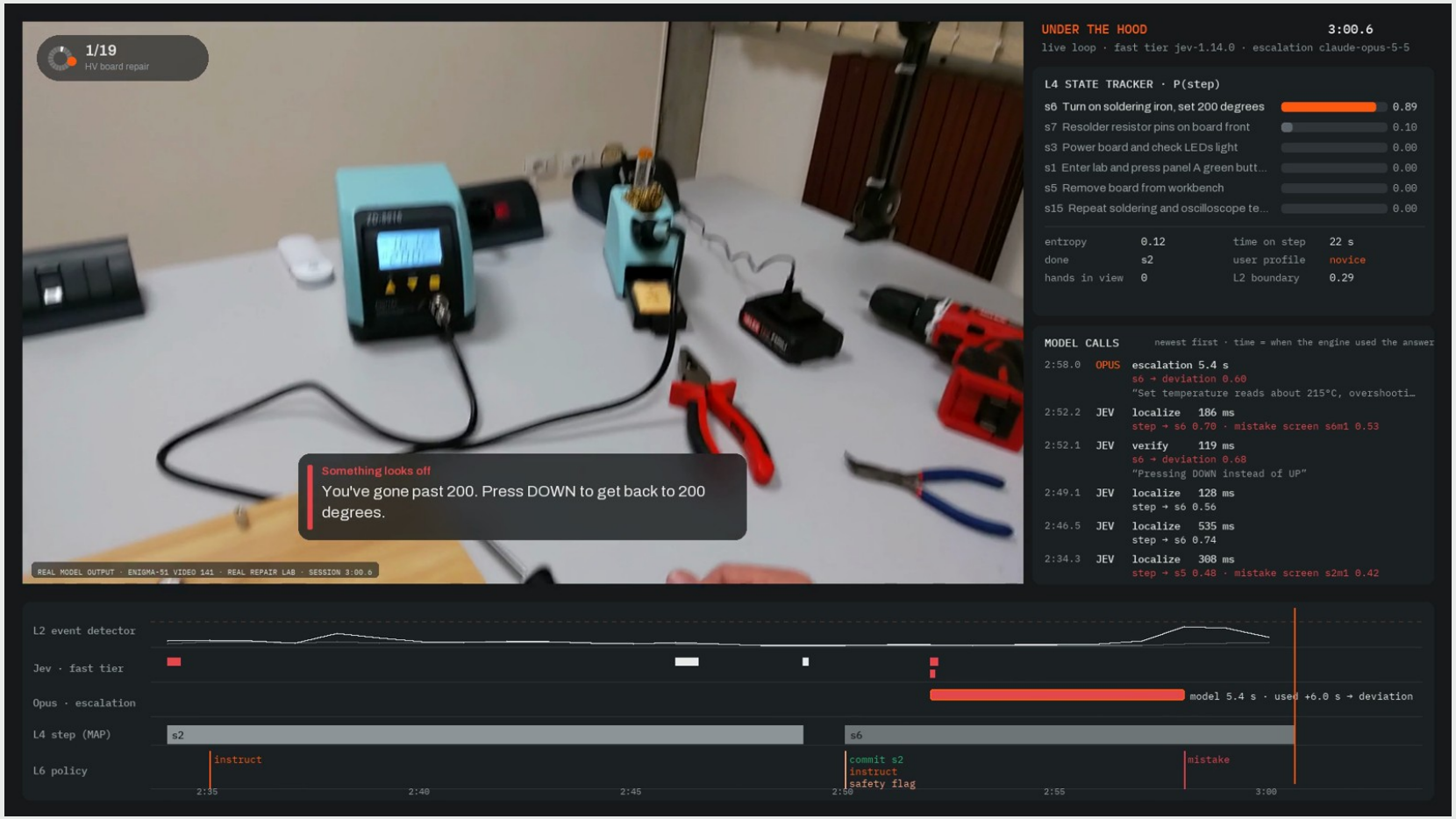
TESTED ONLY ON RECORDINGS IT NEVER SAW · BOTH STILL RISING AT THE LARGEST SIZE WE HAVE

APPENDIX · A4

Full architecture. Orange runs on every frame; the bottom row only when something happens.



Under the hood: the same live session, layer by layer



State tracker. Its belief over the 19 steps, the time on the current step, and whether the tech looks like a novice.

Model calls. Jev answers in 150 ms (median, 130 calls). Opus double-checks 14 moments, about 4 s each.

Timeline. The event detector, both model tiers, the tracked step and every decision to speak or stay quiet.

Cost. $0.165 for the whole 16:40 session, with 0 errors.

A shop uploads the manuals it already has. Praxis turns each procedure into a coach

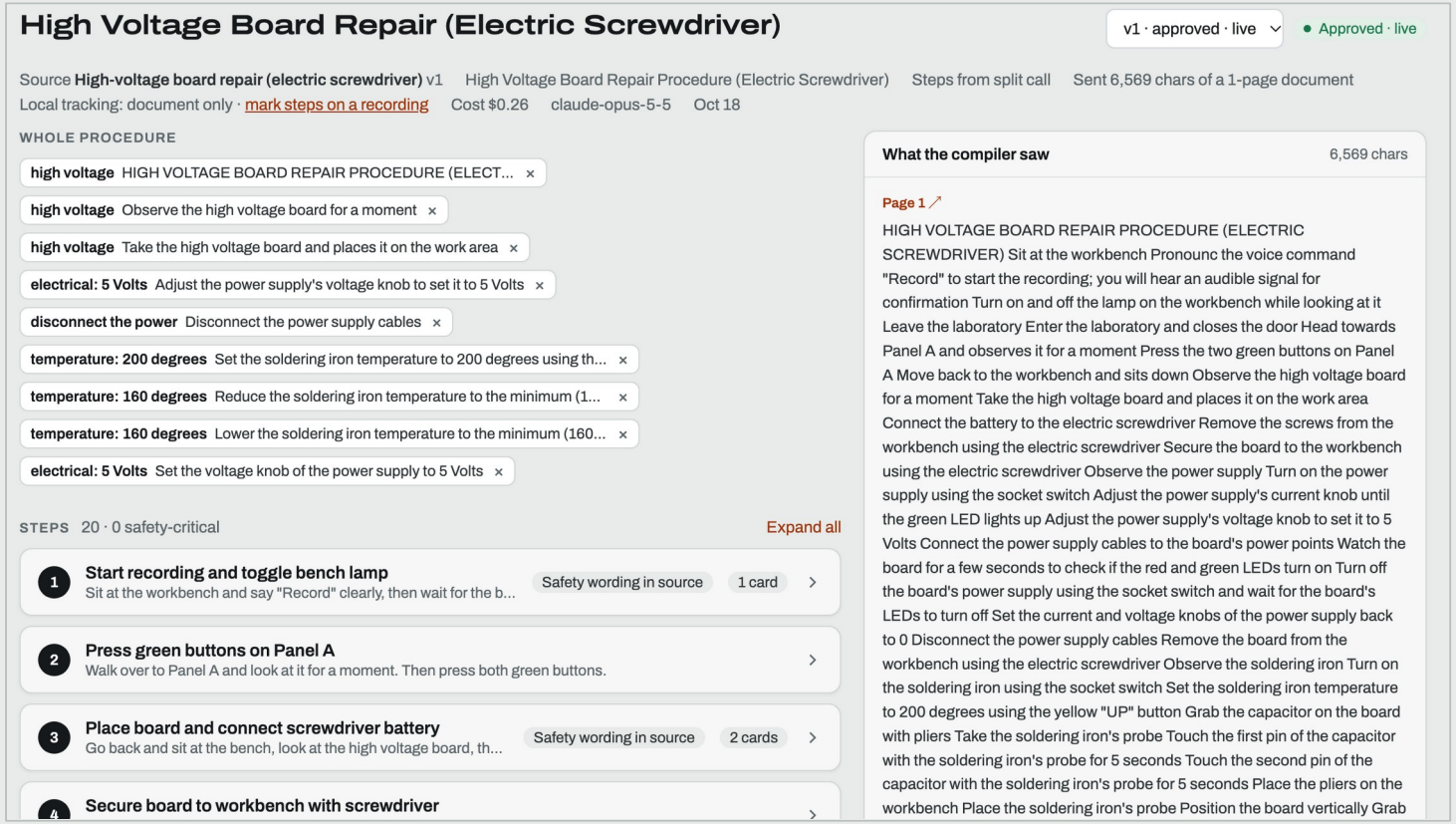
3 DOCUMENTS

DOCUMENT	PAGES	SECTIONS	LIKELY PROCEDURES
FIST 1-1 · Hazardous Energy Control Program FIST_1-1.pdf · v1	97	131	4
FIST 4-1A · Maintenance Scheduling for Mechanical Equipment FIST_4-1A_2020.pdf · v1	222	234	3
FIST 3-8 · Protective Relays: Operation, Maintenance and Field Tests FIST_3-8_2021.pdf · v1	125	97	0

1. **Upload the manuals**
   A 97-page safety manual is indexed in 0.34 s on a laptop, for $0. Its procedures are found automatically.

2. **Review the coached steps**
   Each procedure compiles for about $0.19, with cards quoted from the manual. A manager approves it first.

3. **Coach through smart glasses**
   Live tracking and voice control. Meta and Viture adapters are built, not yet tested on a device.



APPENDIX · A7

What we train in-house, and why a big AI lab won't build this

PART	WHO	WHAT IT DOES
Event detector	OURS · TRAINED	A small temporal network that spots the moment a step ends
Task recognizer	OURS · TRAINED	Works out which procedure the tech is doing
State tracker	OURS · TRAINED	A Bayesian model of where the tech is in the procedure, with step models per procedure
Local checker	OURS · TRAINED	Learned from the frontier model; answers routine "is this step done?" checks for free
Perception	OPEN MODELS	SigLIP2, V-JEPA 2 and hand tracking turn each frame into features. Not trained by us
Verifier	RENTED	A frontier model answers hard questions on 14% of frames. Swappable in one line

WHY A BIG LAB WON'T

They sell one general model. Ours wraps any model, so their best one becomes our swappable verifier.

The hard part is data they don't have: first-person video of skilled work, lined up to the procedure, with verified steps.

And a business they don't run: each shop's manuals and records, and a pilot on site.

APPENDIX · A8

Every approach, honestly: what it does, and how we differ

APPROACH	EXAMPLES	WHAT IT DOES	HOW WE DIFFER
Shadowing	A senior tech	The real teacher today	About $44 an hour of senior time, one new tech at a time
Manuals + videos	PDFs, binders, YouTube	Explain what should happen	Can't see what the tech is doing
Procedure software	MaintainX, Tulip, Augmentir, Dozuki	Steps on a screen, sign-offs, photos; some compile instructions	A checkbox isn't proof. Natural integration partners
Fixed cameras	Retrocausal, Invisible AI, Tulip Vision	Watch one station; record, label and train per station	Ahead of us on live mistake detection at a station; can't follow a tech around a machine
General assistants	Meta AI, Gemini Live	Free help about what's in view	No procedure, step state, calibration or records
First-person, high-touch	Strivr	Per-customer capture and model (vendor docs); early access	No published accuracy; we start from the manuals a shop has
First-person, hardware-led	Viture Helix + NVIDIA XR AI	Real-time SOP coaching on its own safety glasses; ships Q1 2027, from $600	Closest. How SOPs get in and accuracy aren't published. We'd rather run on it
Toolkits	NVIDIA XR AI, Meta's toolkit, Android XR	Plumbing for apps	Complements: what we build on

Market math, and the assumptions behind it

	WHO (BLS)	MATH	PER YEAR
TAM	Installation, maintenance + repair 6,086,190 · electricians 757,220 · production 8,565,760 = 15.4M	15.4M × $50 a month × 12	≈ $9.2B
SAM	General maintenance + repair, machinery mechanics + millwrights, HVAC = 2.47M jobs, ~255k openings a year	255k × $200 × 6 months = $306M; 2.47M × $50 × 12 = $1.48B	≈ $1.8B
SOM (year 3)	1.5% of new maintenance techs ≈ 3,750 new techs a year, ~150 mid-size teams	3,750 × $200 × 6 months	≈ $4.5M ARR

Prices are hypotheses. $200 a month for coaching anchors on Tulip's $100–250 per station; $50 a month for every-worker records is a guess

Openings include people changing jobs, so new-tech counts are an upper bound

Not counted: other countries, field service outside these groups, aviation maintenance, and the robotics upside

APPENDIX · A10

Unit economics, and the pricing risk

DESIGN	$ / VIDEO-MIN	$ / SEAT-MONTH, 2 H/DAY	MARGIN AT $200
Frontier model every 5 s	$0.085	about $214	negative
Praxis engine today	$0.018	about $45	about 77%
Praxis + our local model	$0.008	about $20	about 90%

Labeled assumptions, not sourced: if a senior tech spends 50% of their time on a new hire for 3 months, that's about $11,400 of their time; at 25% for 12 months, about $22,800. We're asking maintenance managers for their real split.

THE RISK

All-shift use (8 h/day) makes a flat $200 seat $1.19 an hour against our $1.08 cost: about a 9% margin. With the local model, about 60%. So price by coached hours or tiers, and move routine checks to the local model first.

Hardware: build, access, glasses and seats

QUESTION	ANSWER
Why not build our own glasses?	Meta, Google, Samsung and Viture + NVIDIA are solving optics, batteries, cameras, radios, distribution and SDKs at a scale we can't match. Our value starts after the camera: procedures, step state, verification, when to speak, records.
What if Meta closes access?	The engine doesn't care where frames come from: each device is a thin adapter over one camera-and-audio interface. Meta is one of several families (Android XR, NVIDIA XR AI glasses, Android headsets).
How do glasses and licenses work?	Glasses are shared site equipment: a site needs enough pairs for the techs coached at the same time, not one per employee. Software is a seat per active new tech that moves to the next hire. We don't resell hardware.

ADAPTER STATUS · OCT 20

- **Any camera or upload**
  tested: runs live in real time on recorded video (local layers ~50 ms a frame)
- **Meta (toolkit 1.0)**
  built, not yet tested on a device
- **Viture / NVIDIA XR AI**
  built, not yet tested on a device
- **Android XR**
  planned

APPENDIX · A12

Data rights: the customer owns its footage

QUESTION	DEFAULT ANSWER
Who owns the manuals and footage?	The customer. Praxis gets a limited license to process them to provide the service.
Can Praxis train on proprietary company video?	Only with contractual permission. Improving our shared models needs a separate, explicit opt-in. Without it, a customer's data stays isolated to that customer.
What leaves the site?	The fast layers can run locally; only event keyframes go to a model (about 14% of frames in our evaluation). Keep as little as possible, for as short as possible.
Workers, unions, monitoring rules	Coaching is for the worker; the employer gets records and aggregates. Face and badge blur is next. Each site's workplace, privacy and union rules come first.

THE LINE WE DON'T CROSS

We never assume we own customer video. Shared-model training happens only on data a customer has opted in.

Safety: Praxis prompts and records, never signs off

SITUATION	WHAT PRAXIS DOES
Safety-critical steps (lockout/tagout)	They stay the tech's job. Praxis prompts and records them, and never approves them.
A procedure before anyone uses it	A manager reviews and approves every compiled procedure before a tech sees it, with each card quoted from the manual.
It can't see well enough	It asks for a better view instead of guessing: 77–98% of blurred or blocked checks, never confidently wrong on them.
It isn't sure	Calibrated confidence decides whether to speak; below the bar it stays quiet or asks. Calibration error fell from 0.22 to 0.02 on held-out checks.
Mistake alerts	Our weakest number today: 4 of 17 alerts named a real annotated error. Every pilot measures it before we claim more.

The risks we see, and what we do about each

RISK	WHAT WE DO
Meta or Google ships a general coach	It answers about what's in view. We go deep in maintenance: procedures, step state, records, pilots on site, and we run on their glasses
Viture + NVIDIA win the safety-glasses slot	Run on Helix via NVIDIA's open XR AI library or its app path, rather than fight it
Glasses don't work end to end yet	Adapters are built but untested on a device; testing at a real site is the $2,500 line (appendix A11)
Mistake detection is weak (4 of 17)	Locked procedures, maintenance data and the tech saying "done" all add evidence. Measured in every pilot
Evaluated on cooking, not maintenance	That's why the grant buys ~80 hours of consented maintenance footage first
Depending on model vendors	The verifier is swappable in one config line, and our local student already handles routine checks
Solo founder	The grant tests demand before hiring. First hires come after the first pilot

Why not consumers first? Because the budget and the repeat data are in business

$3.6B

what Autodesk agreed to pay for MaintainX, an app that puts maintenance procedures on a phone.

$99 → $25

what Frontdoor's expert video help for homeowners fell to, per year, from 2023 to 2024.

The same checklist runs on every machine, every month. Home tasks don't repeat, and the platforms give consumer help away free. Consumers come last.

Founder: what I've built, in more detail

LipTrac. Reads lip movements in real time from ordinary video. Conv3D + Bi-GRU, 60,000+ video frames, tested with 12 participants, provisional patent filed. The hard part was real people under changing lighting and camera angles, while keeping temporal context.

Robotics. Seven years of competitive robotics, where I learned how hard it is to mentor someone without seeing exactly what they see.

Aerial vision. Autonomous path-planning systems for oil-cleanup boats, using aerial imagery.

Praxis, since Oct 17 (130+ commits, solo). A tracking engine with trained event and step models; a 48-clip evaluation harness with a stateless baseline; a company backend that indexes manuals and compiles procedures; a live app that runs on a video stream in real time.

The long game in detail: revenue targets, certifications and robotics

NOW · 2027

Coach new techs

Paid pilots with maintenance teams. With permission, sessions can become labeled training data.

Target: $0.3M · 10 teams

NEXT · 2028

Our own models

Trained on our pilot footage, on the glasses. Cheaper to run and better at maintenance than rented models.

$1.5M · 50 teams · 5×

THEN · 2029

Certifications through Praxis

Partner with NCCER, NIMS and equipment makers so their hands-on tests run on Praxis. Certified techs bring us their next shop.

$4.5M ARR · 150 teams · 3×

LATER · 2030+

Robots: two paths

A Train our own robot models on data customers opted in

B Sell the data and our step checker to robot labs

$13M → $30M ARR, 2030–31

AI LABS ALREADY PAY FOR DATA, AND IT'S GROWING

$2B+ a year

Mercor's run-rate selling expert data to AI labs, doubled in four months

>$100M a year

what robotics companies already spend buying real-world data

$8.6B by 2030

the AI training-data market, growing about 22% a year